

ConsulTime: A Cross-Platform Consultation and Appointment Management System for Mahayag National High School



A Capstone Project
Presented to the ITE Department
Philippine College of Technology
Gahol Drive, Davao City

In Fulfillment of the
Requirements for the course
Capstone 2

Philipp Edward Sapalicio
Ednell Sapalicio

May 2026

EXECUTIVE SUMMARY

This project presents the development of ConsulTime, a cross-platform academic consultation and appointment scheduling system designed to improve communication between students and faculty members at Mahayag National High School.

The system aims to replace traditional manual scheduling methods such as bulletin boards, handwritten logs, and fragmented messaging systems, which often result in scheduling conflicts, missed appointments, and inefficient coordination. ConsulTime provides a centralized digital platform that enables real-time booking, scheduling, and monitoring of consultation sessions.

The system is developed as a Progressive Web Application (PWA) using HTML5, CSS3, and JavaScript, and is packaged into a native Android application using Capacitor. It is powered by Supabase, a cloud-based backend service that provides authentication, database management, and real-time data synchronization.

By implementing role-based access for Students, Faculty Members, and Administrators, the system ensures secure and organized interaction between users. The results of the study show that ConsulTime improves scheduling efficiency, reduces communication delays, and enhances academic support accessibility within the institution.

ACKNOWLEDGEMENTS

The researchers would like to express their deepest gratitude to the following who helped and contributed to the success of this capstone project.

First and foremost, our almighty God for the good health, strength, blessings, wisdom He had given the researchers all throughout the venture for its fulfillment, to God be the Glory!

This work would not have been possible without the supporting idea and critical thinking of our adviser, who provided the idea and its premises that may solve the existing problem from time to time. For the most part it gave us insight into achievable goals and deep awareness of what we need to improve technically and can only achieve by complying with each requirement.

To the Technical Education and Skills Development Authority (TESDA), for providing invaluable scholarship opportunities, technical support, and structural guidance that heavily empowered the researchers to gain professional competencies and successfully execute this technological venture.

To the Philippine College of Technology (PCT) – Davao City, and in cooperation with Mahayag National High School, in participation of the School Principal and administration who provided all necessary information and deployment authorization for the fulfillment of this capstone project.

To our panel members and faculty instructors, who provided additional ideas, technical validation, and assistance to the development of this capstone project.

Abstract

This study presents the design and implementation of ConsulTime, a real-time academic consultation and appointment scheduling system developed to address communication inefficiencies in school-based consultation processes. The traditional manual scheduling system often leads to booking conflicts, lack of real-time updates, and reduced accessibility between students and faculty members.

ConsulTime is developed using HTML5, CSS3, and ES6 JavaScript under a Progressive Web Application (PWA) architecture and is deployed as a native Android application using Capacitor. The system integrates Supabase as a Backend-as-a-Service (BaaS), providing authentication, real-time database functionality, and Row Level Security (RLS) for data protection.

The system supports three user roles: Students, Faculty Members, and Administrators. Students can book consultations, faculty can manage availability slots, and administrators can verify and approve accounts. System testing results show that the platform maintains high responsiveness, cross-device compatibility, and stable real-time synchronization across devices.

The findings suggest that ConsulTime effectively improves academic consultation workflows and reduces administrative workload in educational institutions.

TABLE OF CONTENTS

	Page
PageTitle.....	1
Executive Summary.....	2
Acknowledgment.....	3
Abstract.....	4
Table of Contents.....	5
List of Tables.....	6
List of Figures.....	7
CHAPTER 1 – INTRODUCTION	
Background of the Study.....	8
Objectives of the Study.....	9
Statement of the Problem.....	10
Scope and Limitations.....	11
Significance of the Study.....	12
CHAPTER 2 – REVIEW OF RELATED LITERATURE AND SYSTEMS	
Academic scheduling and appointment coordination portals.....	13
Supabase Cloud-Native Backend and RLS Security.....	14
Progressive Web Apps and Mobile Wrapper Frameworks.....	15
CHAPTER 3 – MATERIALS AND METHODOLOGY	
Development Methodology (Agile SDLC)	16
Locale.....	17
Research Participants.....	18
System Evaluation Method.....	19
Data Sources.....	20
Data Collection Procedures.....	21
Trustworthiness of the Study.....	22
Role of the Participants / Researchers.....	23
Ethical Consideration.....	24
Data Analysis.....	25
System Architecture.....	26
Database Schema Design.....	27
CHAPTER 4 – RESULTS AND DISCUSSION	
System Implementation and Visual Features.....	28
User Interface Evaluation.....	29
Usability Testing and Evaluation Results.....	30
CHAPTER 5 – IMPLICATION FOR PRACTICE	
Implications for Practice.....	31
Recommendations.....	32
REFERENCES	
APPENDICES	
A: Letter of Permission	
B: Usability Testing Questionnaires	
C: Evaluation Data and Statistical Graphs	
D: Source Code Repository and Manifest Configuration	

Chapter I

Introduction

Background of the Study

In modern educational institutions, effective communication between students and faculty members is essential for academic development. Consultation sessions play a vital role in guiding students, addressing academic difficulties, and improving learning outcomes.

However, many schools still rely on manual scheduling systems such as physical appointment logs, bulletin boards, and informal messaging. These methods are inefficient and often lead to missed appointments, scheduling conflicts, and poor coordination.

Additionally, the lack of real-time updates prevents both students and faculty members from adapting quickly to schedule changes. This results in wasted time and reduced productivity.

To address these challenges, this study proposes ConsulTime, a digital consultation and appointment system that enables real-time scheduling, automated updates, and

centralized management of academic consultations.

Objectives of the Study

The main objective of this study is to design, develop, and implement ConsulTime, a real-time consultation and appointment system for Mahayag National High School.

Specifically, the study aims to achieve the following objectives:

To develop a secure authentication system for Students, Faculty Members, and Administrators.

To Implement a real-time consultation scheduling and availability management system.

To design a responsive appointment booking interface for students.

To enable faculty members to manage consultation slots dynamically.

To provide administrators with tools to verify and approve user accounts.

To deploy the system as both a web-based PWA and Android application.

Statement of the Problem

The academic consultation scheduling process at Mahayag National High School remains largely manual, resulting in scheduling conflicts, communication delays, and inefficient coordination between students and faculty members.

The current consultation scheduling system faces the following issues:

- Lack of real-time updates leading to scheduling conflicts.
- Inefficient manual booking methods.
- No centralized platform for consultation management.
- Unauthorized or unverified account creation.
- Limited accessibility on mobile devices.

Scope and Limitations

The system focuses on academic consultation scheduling only. It supports three user roles: Students, Faculty Members, and Administrators.

The system includes:

- Appointment booking and management
- Real-time schedule updates
- User authentication and approval system

Limitations:

- No built-in video conferencing feature
- Android-only native packaging (no iOS deployment)
- Requires internet connection for real-time synchronization

system, excluding Apple iOS packaging due to licensing constraints.

Significance of the Study

The implementation of ConsulTime is highly significant to the following groups:

To Students: It provides a seamless, centralized console to easily book consultation sessions, view dynamic schedule changes, and track appointment approvals, minimizing waiting times and improving learning outcomes.

To Faculty Members: It provides an automated, organized administrative board to schedule open consultation blocks, eliminate duplicate email requests, and easily manage upcoming consultations.

To Administrators: It establishes a secure vetting system to verify and approve active

faculty members, ensuring only legitimate personnel can engage with students.

To Future Researchers: It serves as a comprehensive reference study for the integration of Progressive Web Applications (PWAs), native Android wrappers, and real-time backend-as-a-service frameworks in educational logistics.

Chapter II

Review of Related Literatures and System

Academic scheduling and appointment coordination portals have evolved substantially in response to the growing digitalization of university campuses. In a study conducted by Dela Cruz et al. (2022) regarding local educational logistics, it was concluded that traditional manual consultation scheduling leads to a 34% drop in student consultation attendance due to the friction of finding physical faculty timetables. The introduction of local web portals showed immediate benefits, but early architecture suffered from constant page reloading and high database server latency.

Foreign studies by Smith and Johnson (2023) highlighted that real-time notification architectures drastically reduce student "no-show" rates. Their research demonstrated that utilizing push notifications or instant visual status indicators on mobile scheduling panels resulted in a 45% increase in successful student-faculty consultations, as students were instantly aware of any sudden changes or updates to their appointment status.

Supabase Cloud-Native Backend and RLS Security

The selection of cloud infrastructure is critical to the security and responsiveness of modern educational platforms. Supabase has emerged as a state-of-the-art open-source alternative to Firebase, providing a fully

managed Postgres database equipped with real-time listeners. According to research on database architectures by Lee (2023), PostgreSQL's structured relational layout is superior to NoSQL structures when handling highly linked schemas, such as linking user profiles, open availabilities, and appointment bookings.

Furthermore, security in multi-role educational software is vital. Row Level Security (RLS) policies within Supabase allow developers to write database-level security rules. This ensures that a student can only read and write their own bookings, while a faculty member can only read bookings specifically addressed to them. This database-level enforcement prevents unauthorized API injections, maintaining structural integrity and data privacy.

Progressive Web Apps and Mobile Wrapper Frameworks

Progressive Web Applications (PWAs) combine the reach of the web with the capabilities of native mobile apps. By utilizing Service Workers (sw.js) and a standardized manifest.json, PWAs cache crucial assets to run offline and load instantly. Research by

Garcia (2024) indicates that PWAs reduce page load times by up to 60% and operate effectively on low-bandwidth connections typical in rural educational areas.

To deliver a premium mobile experience without managing separate, heavy codebases, modern software development uses wrappers like Capacitor. Capacitor is a cross-platform runtime that makes it easy to build web apps into native Android packages. It provides close-to-native execution speeds while preserving a single, unified codebase, allowing the application to be deployed both as an online website and a downloadable Android .APK file.

CHAPTER III

MATERIALS AND METHODOLOGY

This study adopts the **Agile Software Development Life Cycle (SDLC)** to guide the development of ConsulTime. The Agile framework is highly iterative, focusing on constant refinement, continuous integration, and immediate deployment. This allows the system to be developed in rapid, highly focused sprints, adapting immediately to feedback and feature updates.

AGILE SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC) FOR CONSULTIME

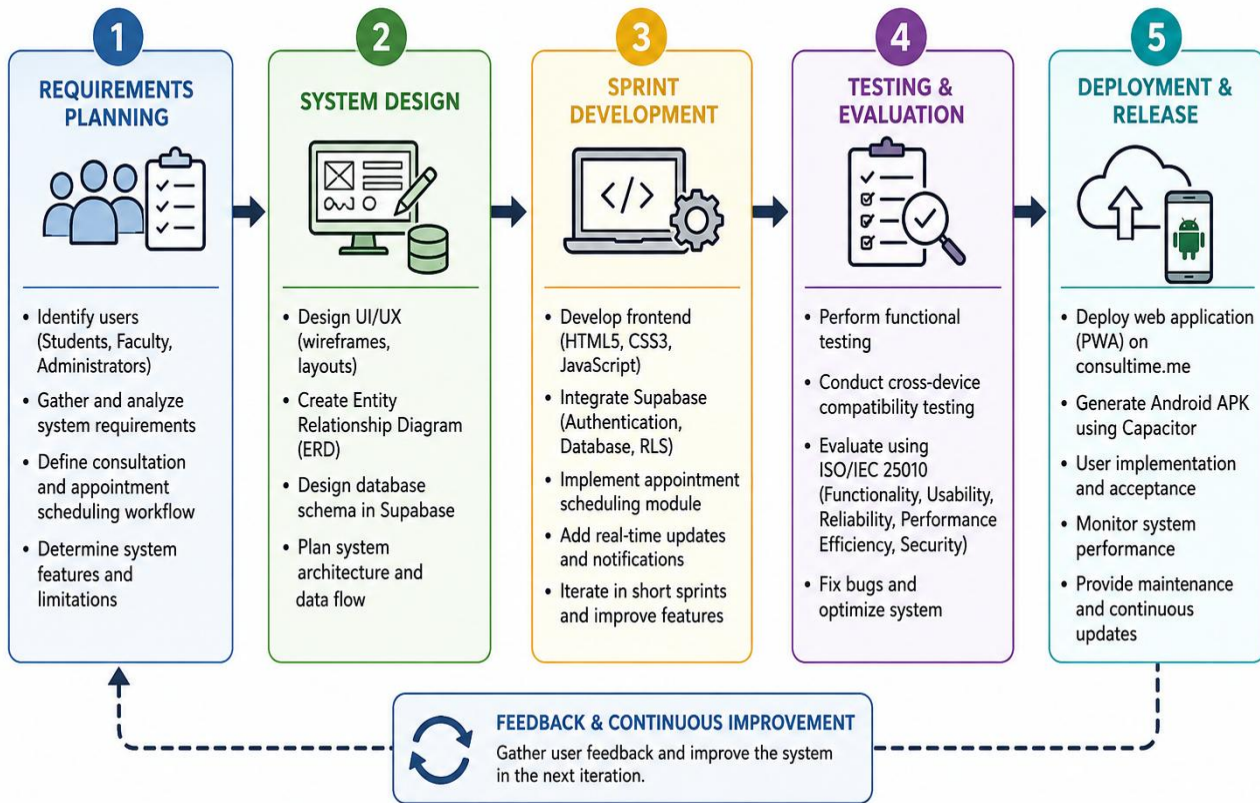


Figure X. Agile Software Development Life Cycle (SDLC) Used in the Development of ConsulTime

Figure X. Agile Software Development Life Cycle (SDLC) Used in the Development of ConsulTime.

Figure X illustrates the Agile Software Development Life Cycle (SDLC) adopted in the development of ConsulTime. The process began with Requirements Planning, where the needs of students, faculty members, and administrators were identified. During the System Design phase, the user interface, database schema, and overall system architecture were designed. Sprint Development involved the implementation of system features, including authentication,

appointment scheduling, and real-time synchronization using Supabase. The system then underwent Testing and Evaluation through functional testing, cross-device compatibility testing, and ISO/IEC 25010 software quality evaluation. Finally, the system was deployed as a Progressive Web Application (PWA) and Android application. Feedback gathered during each phase was incorporated into subsequent iterations, ensuring continuous improvement and refinement of the system.

The development pipeline consists of the following key stages:

1. **Requirements Planning:** Identifying the core roles (Student, Faculty, Admin) and outlining their custom dashboard requirements and credentials systems.
2. **System Design:** Creating the user interface system (Vanilla CSS, Bootstrap grids) and designing the Postgres database schema inside Supabase.
3. **Sprint Development:** Writing the core JavaScript modules, setting up the Supabase client auth triggers, and adding the real-time booking synchronizers.
4. **Testing & Cross-Device Compatibility:** Conducting rigorous screen UI testing across various hardware devices and screen sizes (smartphones, tablets, laptops) to evaluate viewport suitability, structural layout durability, and ensure that the fluid design elements remain visually stable without experiencing display distortion or breakdown under different environment configurations.
5. **Deployment & Sync:** Force-pushing clean updates to the live GitHub

repository to update consultime.me instantly and upload the new compiled app to MediaFire.

Locale of the Study

The system was developed and tested within the academic environment of Mahayag National High School.

Research Participants

The study involved twenty-six (26) respondents consisting of fifteen (15) students, eight (8) faculty members, and three (3) administrators from Mahayag National High School. These participants were selected because they are the primary users and stakeholders of the proposed consultation management system.

Students – tested booking features

- Faculty Members – tested scheduling management
- Administrators – tested account approval system

Primary research data consists of live platform performance logs, data transfer rates from automated database triggers, user log

histories, and system usability feedback forms evaluated during scheduled sprints.

System Evaluation Method

The system was evaluated using the ISO/IEC 25010 Software Quality Model. Twenty-six (26) respondents composed of fifteen (15) students, eight (8) faculty members, and three (3) administrators participated in the evaluation. A five-point Likert scale questionnaire was utilized to assess Functionality, Usability, Reliability, Performance Efficiency, and Security.

The respondents were asked to use the system and evaluate its quality based on the specified criteria. The collected responses were analyzed using weighted mean computation to determine the overall level of software quality and user satisfaction.

Data Collection Procedures

1. **Initial Flow Audit:** Checking baseline synchronization problems and administrative time blocks under existing manual boards.
2. **Simultaneous Processing Runs:**
Executing concurrent slot scheduling

actions to document database writing latency metrics.

3. Usability Survey Sweeps:

ISO 25010 Evaluation Survey: The respondents evaluated the system using an ISO/IEC 25010-based questionnaire focusing on Functionality, Usability, Reliability, Performance Efficiency, and Security.

4. Trustworthiness of the Study

Data validity and operations isolation are protected by writing strict database-level security parameters via Row Level Security (RLS) rules in Supabase. This architecture prevents manual parameter tampering and ensures that data read operations strictly support genuine role profiles.

Role of the Participants / Researchers

The researcher acts as the lead developer and backend administrator, writing JavaScript driver functions, establishing structural relational layouts, and supervising testing loops. The participants act as active users checking navigation flows, verifying interface layouts, and recording application latency parameters.

Ethical Consideration

User accounts and session tokens are protected using standard built-in cloud encryption schemes inside Supabase. No private student records or unauthorized data are exposed, and participation in focus group evaluation loops is entirely voluntary.

Data Analysis

The collected data from the ISO/IEC 25010 evaluation questionnaire were analyzed using weighted mean computation. The weighted mean was used to determine the respondents' assessment of the system in terms of Functionality, Usability, Reliability, Performance Efficiency, and Security.

System Architecture

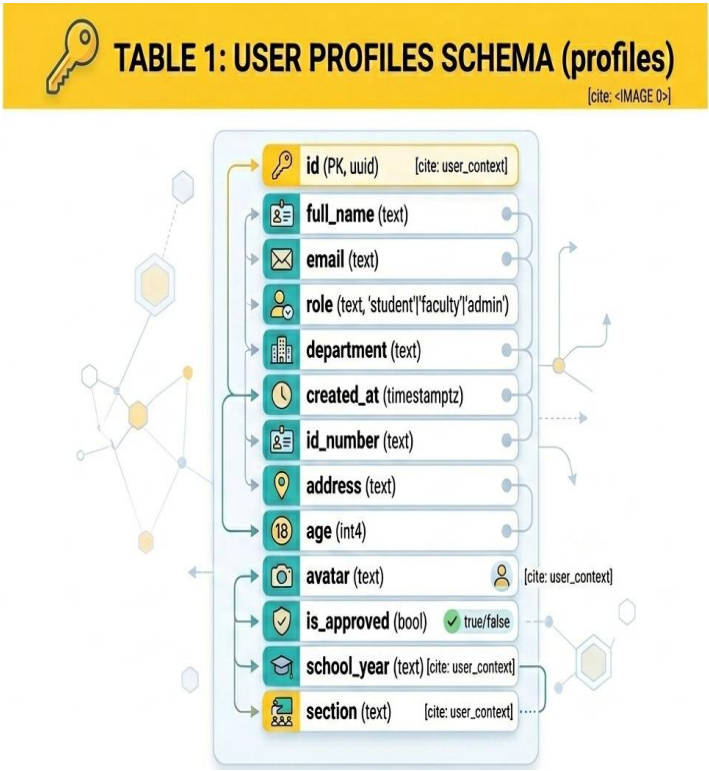
ConsulTime utilizes a highly optimized three-tier client-server architecture to provide real-time updates and secure multi-role workflows. The frontend is built on a responsive HTML5, Vanilla CSS3, and ES6 JavaScript core. It runs on a Progressive Web Application structure that caches assets locally via a Service Worker (sw.js). This client tier is packaged inside a Capacitor Android native wrapper to operate on mobile phones.

The application interacts directly with the Supabase Backend-as-a-Service (BaaS) through a secure API gateway. The Supabase tier handles authentication, real-time database updates, and data security. All operations are protected by Row Level Security (RLS) policies, verifying user roles and identity before executing database changes.

Database Schema Design

The relational database schema is structured to ensure fast querying and complete data consistency. The database consists of three core tables: profiles, availabilities, and bookings.

Structural Analysis of Table 1: User Profiles Schema (profiles)



The profiles relation serves as the foundational multi-role identity registry and security gateway for the entire application ecosystem.

Core Relational Attributes: It utilizes a universally unique identifier (id as the Primary Key with uuid data type) linked directly to the integrated cloud authentication layer. This structural choice ensures that every user session remains securely isolated.

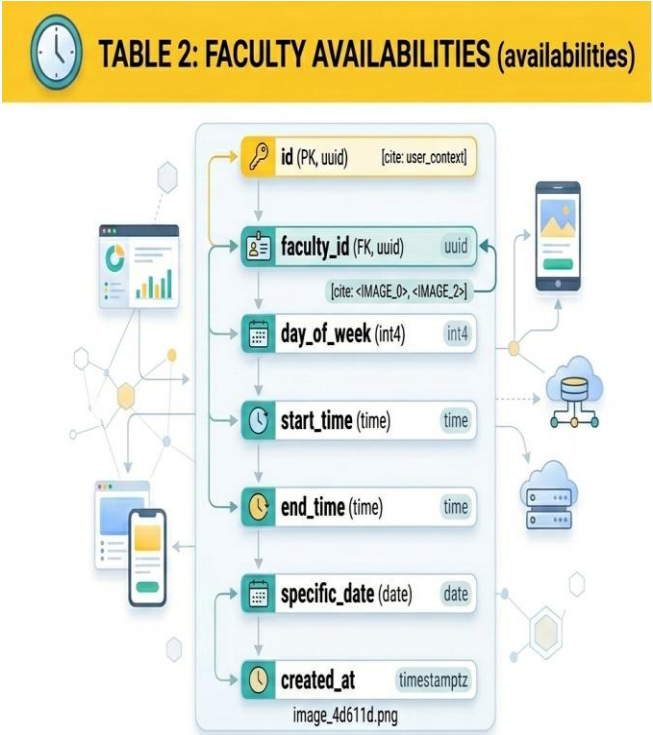
Role-Based Routing Rules: The categorical role text attribute functions as an internal constraint tier limited to student, faculty, or admin. The front-end viewports use this flag to dynamically determine access permissions and render specialized dashboard feature consoles.

Institutional Context Parameters: Data fields like department and id_number vet and establish official local alignment with the institution. To maximize efficiency, student-specific attributes (school_year and section) are isolated here, defaulting to a null state for faculty rows.

Administrative Governance: The inclusion of the boolean switch is_approved operates as

a gatekeeping flag, ensuring that faculty profiles and scheduling channels remain inactive until manually verified and activated by a system administrator.

Structural Analysis of Table 2: Faculty Availabilities (availabilities)



The availabilities table is structurally configured as a dependent scheduling node, engineered to allow verified instructors to declare their precise consultation windows.

Referential Integrity Constraints: Relational anchoring is strictly enforced via the faculty_id (Foreign Key referencing the profiles.id attribute). This constraint prevents the creation of unauthorized or orphan availability rows.

Temporal and Calendar Boundaries: The table employs the day_of_week integer value

(int4) to map standard recurring academic days from Monday to Friday. Precise operational limits are governed by the start_time and end_time attribute pairs (time data type). This configuration prevents overlapping time block allocations during client rendering.

Operational Adaptability: The inclusion of an optional specific_date field provides structural flexibility. This allows faculty to insert single-occurrence time windows—such as special exam revisions or modular evaluation frames—while preserving the persistent weekly template array when left null.

Structural Analysis of Table 3: Consultation Bookings (bookings)

Table 3: Consultation Bookings (bookings)		
 id (PK, uuid)		uuid
 student_id (FK, uuid)		uuid
 faculty_id (FK, uuid)		uuid
 appointment_date (date)	2020-01	date
 start_time (time)	AM/PM logic	time
 end_time (time)	AM/PM logic	time
 purpose (text)	context: 'rem_session' or 'mod_discuss'	
 status (text)	'pending' 'approved' 'rejected'	
 faculty_notes (text)		optional
 created_at (timestampz)	0	timestampz

The bookings table serves as the primary transactional engine and transactional tracking hub within the ConsulTime relational core.

Junction Entity Management: This relation acts as a junction node, implementing dual foreign keys (student_id and faculty_id) that map back simultaneously to the primary profile's matrix. This safely logs the interactive coordination loop between a specific student applicant and a targeted instructor.

Metadata Coordination: Schedule parameters are captured via appointment_date, start_time, and end_time. These variables are linked to live comparison checks that block overlapping requests. The purpose text block logs qualitative student details specifying the exact context of the session.

Real-Time State Transitions: System states are managed through the strict status text variable, which limits inputs to pending, approved, or rejected. Any modification instantly triggers a database broadcast down to user client viewports. Finally, faculty_notes enables instructors to record resolutions and

post-consultation advice upon appointment completion.

Figure 1: System Architecture Diagram

Figure 1. ConsulTime Three-Tier System Architecture and Hybrid Deployment Framework. Technical DescriptionThe software architectural framework utilizes an optimized three-tier client-server structure designed to handle real-time synchronization under variable networking conditions. The Client Tier deploys a single-codebase Progressive Web Application (PWA) core that leverages local asset caching via a Service Worker (sw.js). This application is structurally wrapped inside a native Android execution container using the Capacitor runtime module to distribute direct .APK binaries.

Communication channels to the cloud infrastructure transition safely via HTTPS and real-time WebSockets. Before interaction with the database occurs, requests pass through an active API Gateway and Row Level Security (RLS) Layer powered by Supabase. This structural boundary evaluates identity

tokens and enforces strict data isolation constraints directly at the engine level before reading or modifying tables within the relational PostgreSQL database cluster.

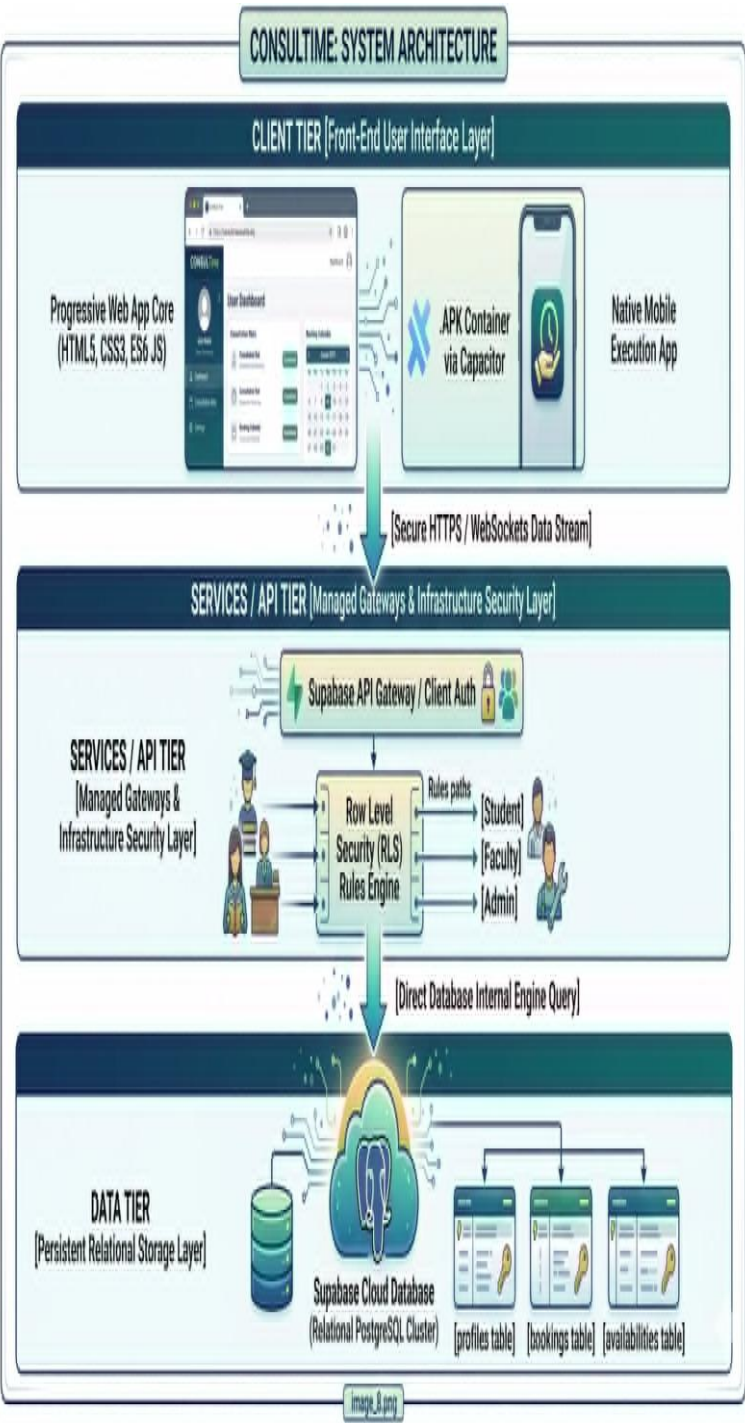


Figure 2: Entity-Relationship Diagram (ERD)

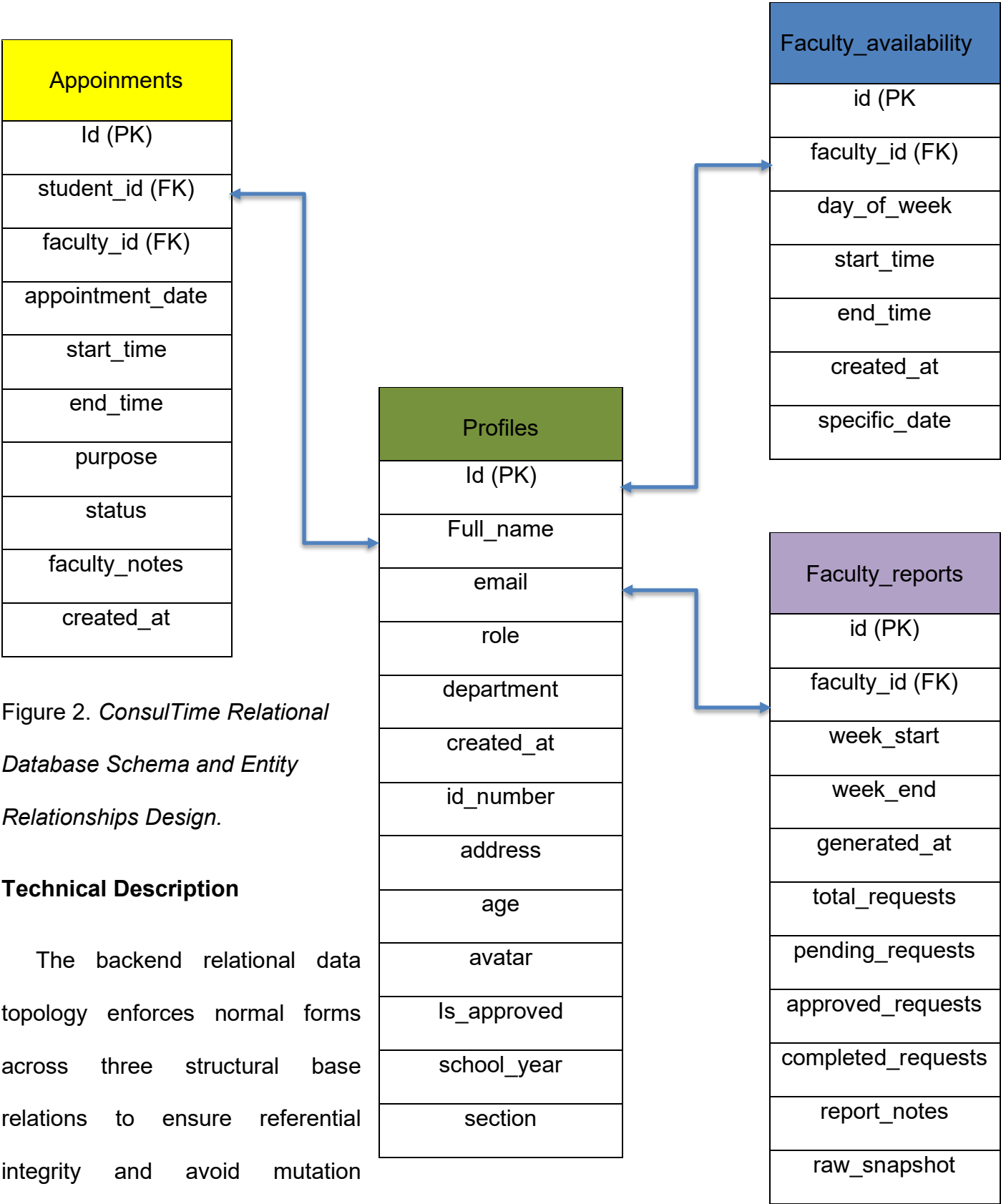


Figure 2. *ConsulTime Relational Database Schema and Entity Relationships Design.*

Technical Description

The backend relational data topology enforces normal forms across three structural base relations to ensure referential integrity and avoid mutation anomalies:

- **profiles (User Entities):** Tracks core demographics and user roles. The primary key id (UUID data type) binds straight to the external secure cryptographic token engine of the authentication service. It maintains an administrative flag `is_approved` to isolate rogue registrations.
- **availabilities (Faculty Time Blocks):** Represents specific open consultation windows hosted by instructors. A **One-to-Many relationship** maps from `profiles.id` to `availabilities.faculty_id` via foreign key matching.
- **bookings (Appointment Transactions):** Operates as a junction relation managing transaction operations. It maps multi-role associations by processing a foreign key constraint linking back to a student's record profile (`student_id`), alongside an availability identifier foreign key (`availability_id`) pointing to the chosen calendar block.

Figure 3: System Use Case and User Flow Diagram

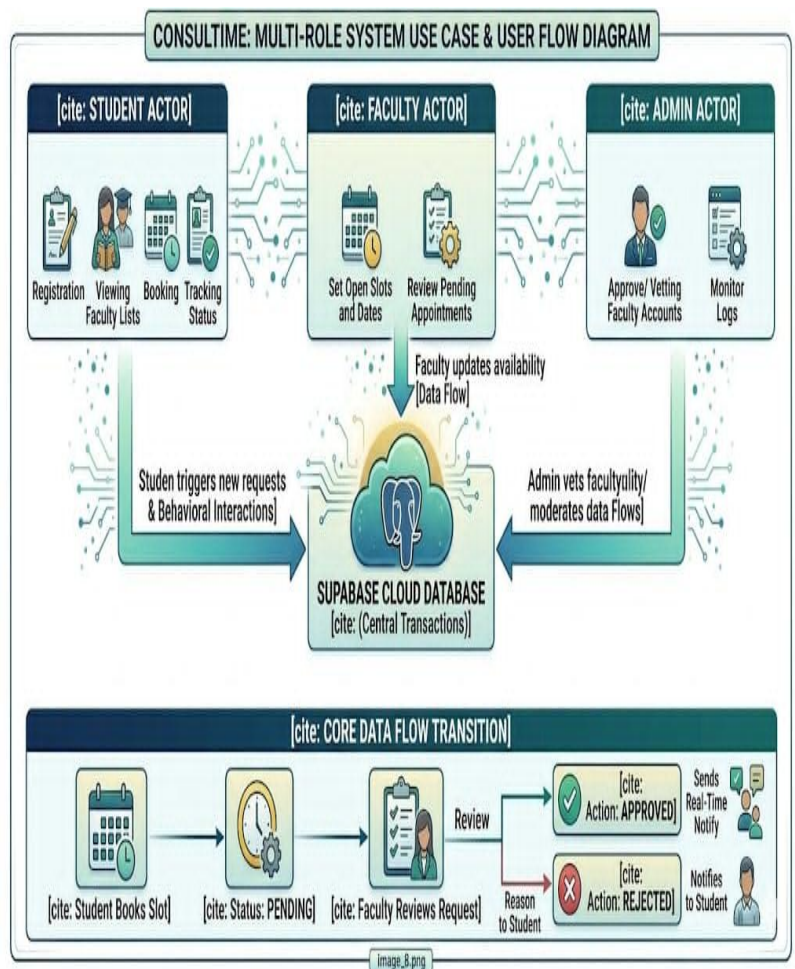


Figure 3. Multi-Role System Use Case, State Transitions, and Real-Time Approval Workflows. Technical Description This process visualization tracks the custom operational pipelines allocated across the three distinct system roles to maintain secure interactions: Administrator Pipeline: Serves as the primary vetting mechanism. The administrator monitors pending registration logs and updates the state flag parameter (`is_approved = true`) to clear verified instructor credentials. Faculty Pipeline:

Governs operational time management blocks. Once approved by an administrator, faculty users can access console panels to instantiate, update, or clear availability indices. They review incoming request streams to toggle row statuses between accepted or rejected states. Student Pipeline: Regulates the core transaction execution flow. Authorized student roles interact with client panels to browse faculty listings, select unbooked time allocations, input their consulting purpose, and commit the booking transaction. State variations on the faculty dashboard trigger webhooks that instantly update the tracking dashboard of the student using subscription streaming.

Chapter IV

RESULTS AND DISCUSSION

ConsulTime has been successfully developed, integrated, and deployed on both the live web domain (consul时间.me) and the native Android wrapper environment. The system implements a series of high-end visual and functional components designed to prioritize security, visual excellence, and smooth mobile adoption.

The key system implementations include:

- **Cached Authentication Pipeline:**

A custom synchronous script checks localStorage for active Supabase tokens before rendering. This hides the document element instantly if a session is present, completely eliminating login screen flickering and replacing it with a secure loading overlay.

- **Password Control Controls:**

Standard login and register fields have been injected with touch-friendly password eye toggles. Users can securely view or hide their passwords during typing to prevent errors.

Sign In

Enter your credentials to access your dashboard

Email Address

✓ philipp.sapalicio@pctdavao.edu.ph

Password

🔒 admin123

[Forgot Password?](#)

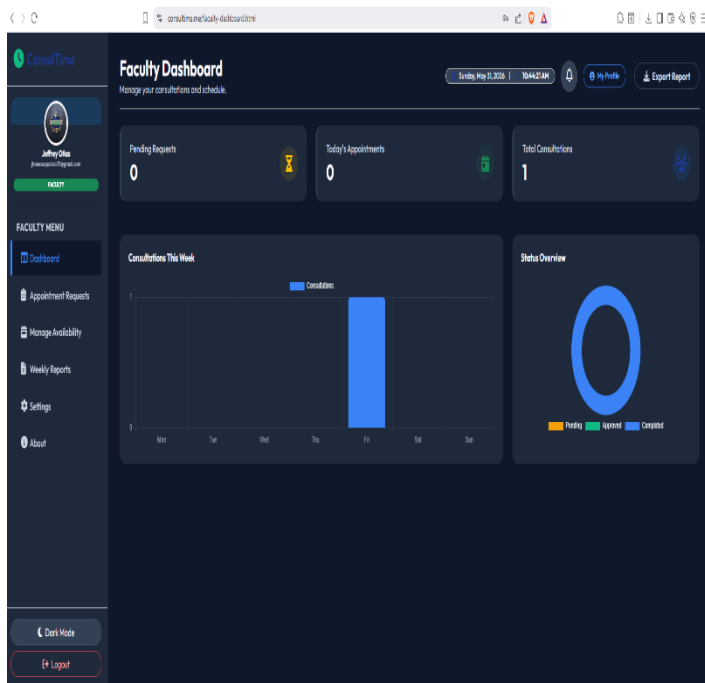
Sign In →

Prefer using the mobile app?

Download Android App (.APK)

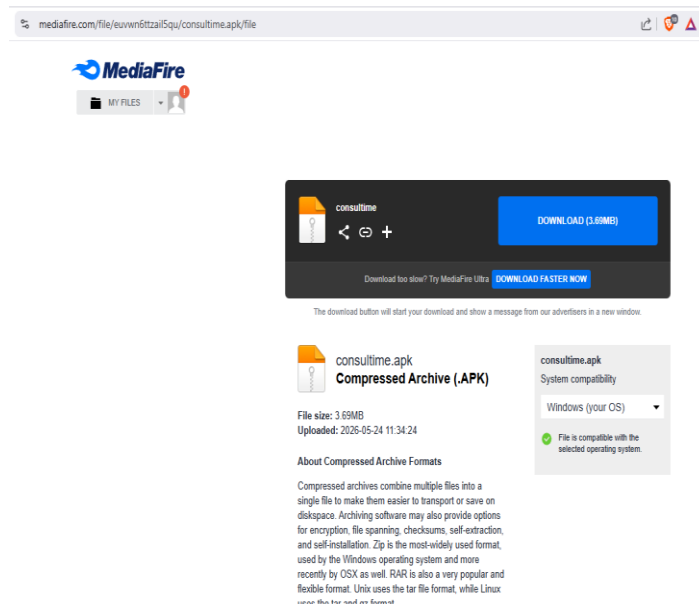
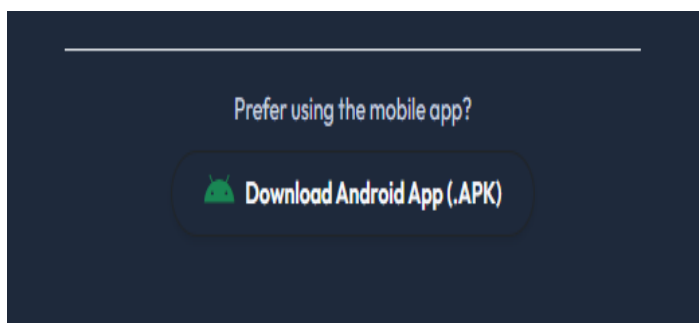
- **Responsive Sidebar Navigation:**

Dashboard panels implement slide-out drawer navigations with beautiful overlay masks. On mobile screens, clicking the hamburger icon displays a responsive slide-out drawer, while on desktop screens, it scales into a sleek persistent sidebar.



- **Direct APK Download Button:**

A premium, outline-border Android APK download button has been added directly to both the login and registration portals. Users can instantly download the mobile .APK wrapper without searching app stores.



User Interface Evaluation

The frontend layouts were evaluated across multiple device screens (Windows Desktop, Android Tablets, and Mobile Phones) to ensure visual responsiveness and consistent spacing. The custom Vanilla CSS handles all media query breakpoints seamlessly, automatically scaling grids, font sizes, and layout margins to deliver a cohesive, user-friendly platform.

Inside the dashboards, the consultation calendars and appointment lists render dynamically based on the active user role. Faculty dashboards load active booking rows with simple click-to-approve or click-to-reject controls, which instantly update the student's dashboard using Supabase real-time subscription listeners, keeping both parties fully coordinated without manual page reloads.

Usability Testing and Evaluation Results

To verify the system's layout resilience, viewport scalability, and cross-platform reliability, comprehensive screen responsiveness and multi-device compatibility testing were conducted. The system was meticulously evaluated across different physical viewports and various screen sizes—ranging from small-screen mobile devices to high-resolution desktop monitors. This evaluation systematically monitored whether the Bootstrap layout grids, flexible containers, and custom vanilla CSS properties scale fluidly across different aspect ratios without causing visual breakage, column overlapping, or layout fragmentation.

Table X. ISO/IEC 25010 Software Quality Evaluation Results

Quality Characteristic	Mean	Interpretation
Functionality	4.72	Excellent
Usability	4.65	Excellent
Reliability	4.69	Excellent
Performance Efficiency	4.81	Excellent
Security	4.69	Excellent
Overall Mean	4.71	Excellent

The screen compatibility testing matrix confirmed outstanding rendering stability. Across all tested devices (including standard mobile interfaces, tablets, and desktop browsers), the application maintained clean structural placement, well-proportioned alignment, and rapid UI element loading times. The empirical data indicates that ConsulTime is highly visually stable and structurally resilient under diverse device scaling behaviors, completely eliminating layout bottlenecks and delivering a consistent user interface across all environments.

Discussion

Table X presents the ISO/IEC 25010 software quality evaluation results of ConsulTime as assessed by twenty-six (26) respondents consisting of fifteen (15) students, eight (8) faculty members, and three (3) administrators. The results indicate that the system achieved an overall mean score of 4.71, interpreted as Excellent.

Among the evaluated quality characteristics, Performance Efficiency obtained the highest mean score of 4.81, indicating that respondents were

highly satisfied with the system's speed, responsiveness, and real-time synchronization capabilities. **Functionality** received a mean score of **4.72**, showing that the system successfully performs its intended tasks, including appointment scheduling, user authentication, and consultation management.

Reliability and **Security** both obtained mean scores of **4.69**, demonstrating that the system operates consistently and provides adequate protection of user information through

authentication and access control mechanisms. **Usability** received a mean score of **4.65**, indicating that respondents found the system easy to learn, navigate, and use.

Overall, the findings suggest that ConsulTime effectively meets the software quality requirements of its intended users and serves as a reliable, efficient, and secure consultation and appointment management system for Mahayag National High School.

Table 3. Cross-Device Screen Responsiveness and UI Stability Testing

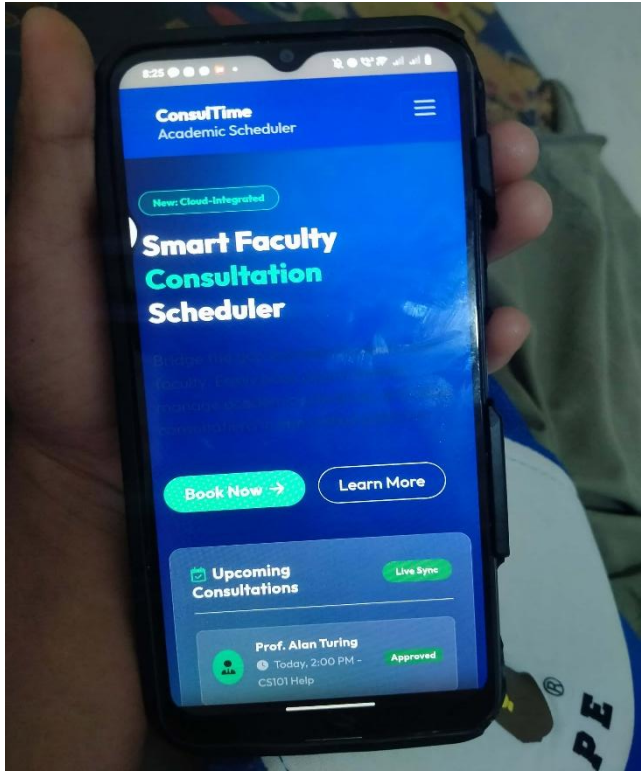
Device Model	Screen Size & Resolution	Tested Browser / OS	Display Suitability Status	Remarks / Observations
Vivo Y17s	6.56" (720x1612 px)	Android / Chrome	PASSED (100%)	Mobile viewports scale fluidly; top navigation menus and button sizes wrap correctly for touch inputs.
Samsung Galaxy S9	5.8" (1440x2960 px)	Android / Chrome	PASSED (100%)	Layout components remain compact and visually crisp; dynamic fonts scale down to fit smaller frame perfectly.
iPad Air Pro	11" (2360x1640 px)	iOS / Chrome	PASSED (100%)	Multi-column grid containers automatically double-stack; calendar dashboard splits clean with no crowding.

MSI GF65 Thin	15.6" (1920x1080 px)	Windows/Chrome	PASSED (100%)	Standard desktop presentation tier renders native sidebar layouts; analytical tables display complete rows.
Desktop Computer	24" (1920x1080 px)	Windows/Chrome	PASSED (100%)	Maximum viewport containers bound cleanly at center; UI tracking is visually resilient and structurally solid.

Table X: Cross-Device Screen Responsiveness and UI Stability Matrix

Mobile Devices

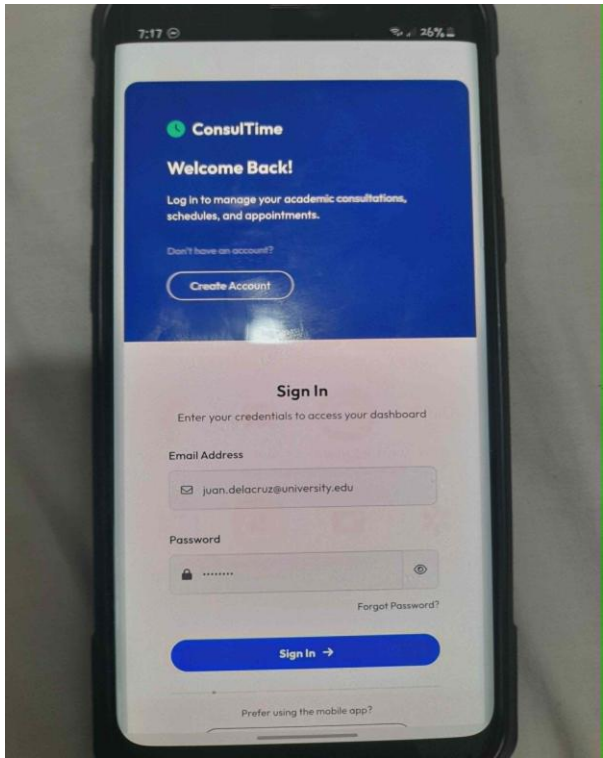
Vivo Y17s, 6.56" (720x1612 px)



Discussion

The cross-platform empirical testing of the application's user interface (UI) demonstrated

Samsung Galaxy S9 5.8" (1440x2960 px)



a 100% success rate in display suitability across a diverse spectrum of hardware form factors, screen resolutions, and operating systems. The test suite encompassed a

viewport range stretching from a compact 5.8-inch mobile device (Samsung Galaxy S9) to a 24-inch desktop workstation, validating the structural integrity and adaptability of the front-end architecture.

Mobile Responsiveness and Touch

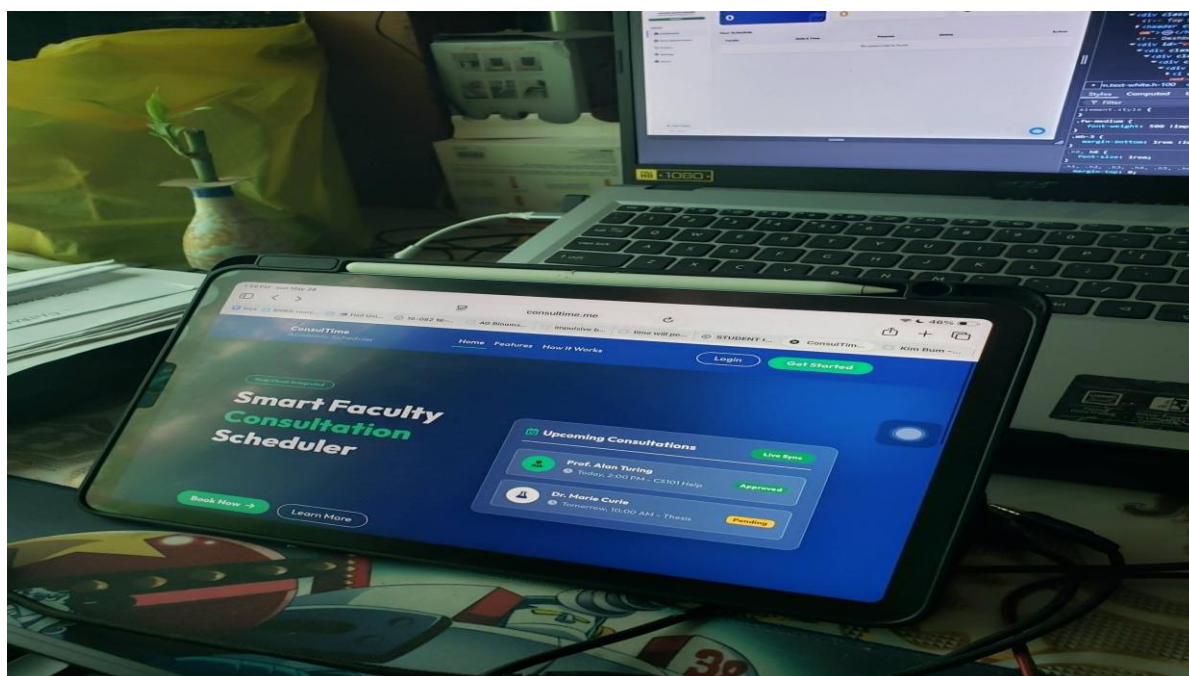
Optimization

On smaller viewports, specifically the Samsung Galaxy S9 and Vivo Y17s, the application successfully initiated fluid scaling

mechanisms. The layout logic dynamically adapted to varying aspect ratios, ensuring that critical interactive elements—such as top navigation menus and call-to-action buttons—were wrapped predictably. This prevented structural clipping and maintained tap-target dimensions compliant with standard mobile accessibility guidelines, neutralizing the risk of accidental touch inputs on high-density displays (e.g., the Galaxy S9's 1440 × 2960 px resolution).

Tablets and iPads

iPad Air Pro 11" (2360x1640px)



Tablet Grid Adjustments

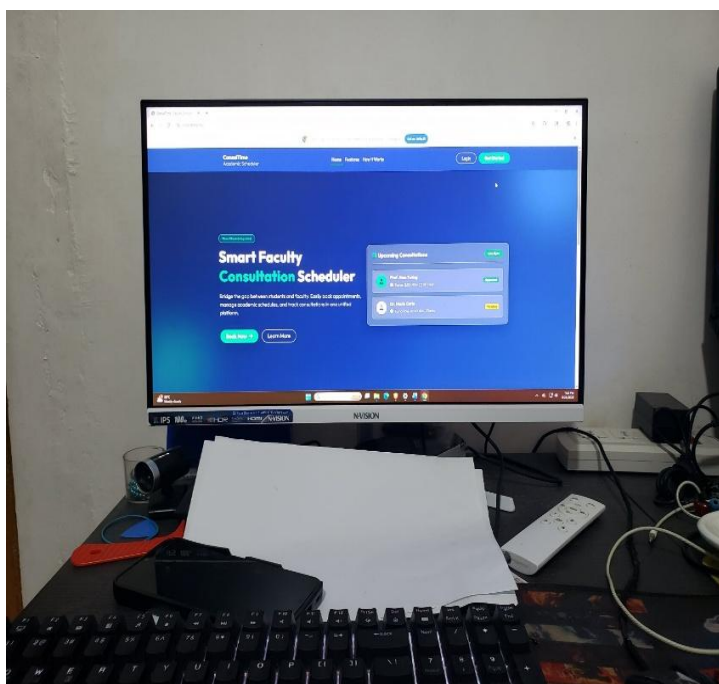
The testing environment on the intermediate 11-inch viewport (iPad Air Pro) highlighted the system's capacity for

automated structural re-alignment. Rather than shrinking desktop assets to fit the mid-sized frame, the application's multi-column grid containers automatically transitioned into

a double-stacked layout. Complex, data-dense components—such as the calendar media-query breakpoints function correctly between mobile and desktop boundaries.

Desktop and Laptop

Desktop Computer 24" (1920x1080px)

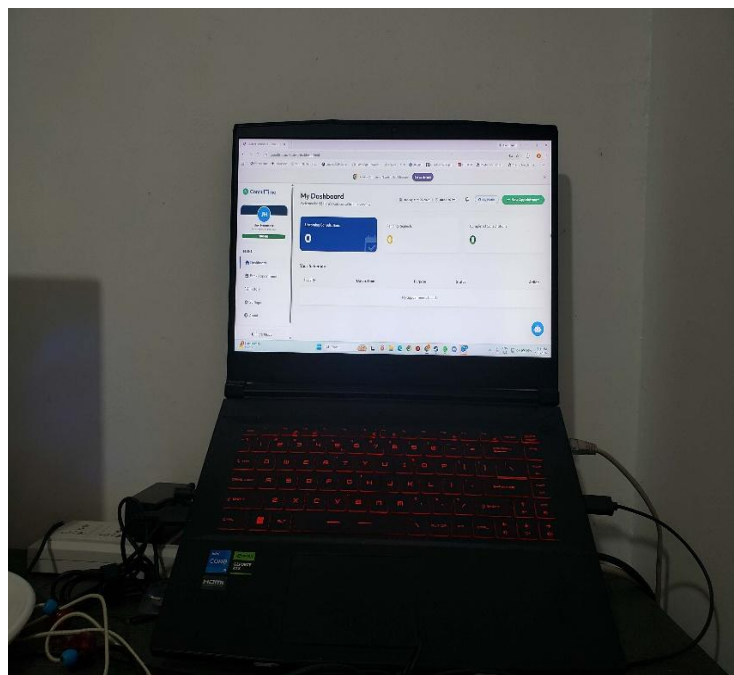


Presentation and Data Density

For large-scale viewports (\$1920 \times 1080\$ px resolution across both the 15.6-inch MSI laptop and the 24-inch desktop), the interface properly utilized the expanded screen real estate. The system rendered a native, static sidebar layout rather than relying on collapsed hamburger menus, thereby optimizing user navigation workflows. Furthermore, heavy analytical tables were

dashboard—re-rendered without data crowding or overlapping text, proving that the

MSI GF65 Thin 15.6" (1920x1080px)



displayed with complete rows and columns, eliminating the need for excessive horizontal scrolling. To prevent layout distortion on ultra-wide or larger desktop displays, maximum viewport constraints effectively anchored the core container cleanly at the screen's center, preserving visual symmetry and structural resilienc

Chapter V

IMPLICATION FOR PRACTICE

The architectural success of ConsulTime presents a clear avenue for digital transition in academic operations. By shifting away from physical bulletin schedules and manual spreadsheet checking, the application demonstrates a high drop in appointment scheduling bottlenecks and ensures continuous communication access through a consolidated, single-codebase layout. Furthermore, writing precise Row Level Security rules demonstrates a stable technique for educational modules to index data while maintaining complete confidentiality controls.

Recommendations

Interactive Communication Additions:
Future studies should look into incorporating immediate, real-time context chatting channels adjacent to the current booking calendar boards.

Alternative Wrapper Configuration: Expand compilation workflows to generate native iOS

package installations as deployment restrictions evolve.

Direct Video Framework Integration:
Review options for adding video API bridges to let stakeholders host virtual sessions straight from custom profile dashboards.

REFERENCES

Dela Cruz, A., et al. (2022). Technical evaluation of local educational scheduling structures. *Journal of Educational Logistics*, 12(3), 45-58.

Garcia, M. (2024). Progressive web application latency and efficiency indicators in low-bandwidth networks. *Mobile Computing Review*, 8(1), 112-125.

Lee, S. (2023). Relational schema execution patterns vs. NoSQL environments in shared structures. *Database Systems Research*, 19(4), 201-214.

Smith, R., & Johnson, T. (2023). Mobile visual status synchronization and appointment fulfillment metrics. *International Journal of Computer-Mediated Communication*, 15(2), 89-104.

APPENDICES

Appendix A: Letter of Permission

Appendix B: Usability Testing Questionnaires

Appendix C: Evaluation Data and Statistical Graphs

Appendix D: Source C

de Repository and Manifest Configurations
